

# N6Cam — Test Report

**Date:** 2026-05-22 19:09:59

**Device:** SIANA N6Cam (STM32N657 / IMX335 / Neural-ART NPU)

**Firmware:** 01.08.2593089169

**MCU UID:** 003031564236500400420024

**Shell TTY:** /dev/ttyACM1

**Runtime:** 34 seconds

**Result:** 3 TEST(S) FAILED — Total: 56 | Pass: 53 | Fail: 3 | Skip: 0

| Test ID | Description                                      | Result |
|---------|--------------------------------------------------|--------|
| T00.1   | Kit CDC port /dev/ttyACM1 exists and is readable | PASS   |
| T00.2   | Shell `help` lists user-defined commands         | PASS   |
| T00.3   | `fw` returns FW VER: ...                         | PASS   |
| T00.4   | `uid` returns MCU UID                            | PASS   |
| T00.5   | `uptime` returns UPTIME ms                       | PASS   |
| T01.1   | `echo query` reports state + ok ack              | PASS   |
| T01.2   | `echo off` succeeds (ok-ack)                     | PASS   |
| T01.3   | After `echo off`, no echoed prompt before reply  | PASS   |
| T01.4   | `echo on` restores echo + ok ack                 | PASS   |
| T02.1   | `rtc` returns DD/MM/YY HH:MM:SS                  | PASS   |
| T02.2   | `rtc set DDMMYYYYHHMMSS` accepts a valid value   | PASS   |
| T02.3   | RTC reads back the value we just set             | PASS   |
| T02.4   | `version` prints Application:                    | PASS   |
| T03.1   | `irled on` reports state=1 + ok-ack              | PASS   |
| T03.2   | `irled off` reports state=0                      | PASS   |
| T03.3   | `motion sense 65 45` accepted, echoed back       | PASS   |
| T03.4   | `motion query` returns persisted 65/45           | PASS   |
| T04.1   | `img quality 75` set + ok-ack                    | PASS   |

|       |                                                                            |                                              |
|-------|----------------------------------------------------------------------------|----------------------------------------------|
| T04.2 | `img color RGB` accepted                                                   | PASS                                         |
| T04.3 | `img chroma 1` accepted                                                    | PASS                                         |
| T04.4 | `img query` shows quality=75 color=RGB chroma=1                            | PASS                                         |
| T05.1 | `detect start` succeeds                                                    | PASS                                         |
| T05.2 | `detect profile 3 1` (people+vehicles, save-SD)                            | PASS                                         |
| T05.3 | `detect profile query` returns 0x03 / 0x01                                 | PASS                                         |
| T06.1 | `notify enable 0x10` (people-detect bit) accepted                          | PASS                                         |
| T06.2 | `notify period 30` accepted                                                | PASS                                         |
| T06.3 | `notify query` shows mask=0x10 period=30                                   | PASS                                         |
| T06.4 | `notify trigger 16` emits +SDVRNTF JSON with rsn=16                        | PASS                                         |
| T06.5 | `notify disable` succeeds                                                  | PASS                                         |
| T07.1 | `sd query` reports mounted                                                 | PASS                                         |
| T07.2 | `sd ls` returns at least one entry                                         | PASS                                         |
| T07.3 | `photo saved` returns SoW §7 filename<br>4325412_22052026_190930.rdy       | PASS                                         |
| T07.4 | New .rdy file appears on SD after `photo saved`                            | FAIL —<br>not seen<br>in sd ls<br>within 15s |
| T08.1 | `frame clear` reverts to live camera                                       | PASS                                         |
| T08.2 | Upload a synthetic 192x192 grey frame                                      | PASS                                         |
| T08.3 | `frame query` reports loaded                                               | PASS                                         |
| T08.4 | `frame run` returns a count + NN time on a grey frame<br>count=0 NN=16.8ms | PASS                                         |
| T09.1 | astronaut.jpg (real person) → NN detects ≥1 person<br>count=1 NN=16.9ms    | PASS                                         |
| T09.2 | camera.jpg (real person) → NN detects ≥1 person<br>count=1 NN=16.9ms       | PASS                                         |
| T09.3 | cat.jpg (no person) → NN returns 0 detections<br>count=0 NN=16.9ms         | PASS                                         |
| T09.4 | coffee.jpg (no person) → NN returns 0 detections<br>count=0 NN=16.9ms      | PASS                                         |

|       |                                                                                                            |                         |
|-------|------------------------------------------------------------------------------------------------------------|-------------------------|
| T09.5 | moon.jpg (no person) → NN returns 0 detections<br>count=0 NN=16.9ms                                        | PASS                    |
| T09.6 | rocket.jpg (no person) → NN returns 0 detections<br>count=0 NN=16.9ms                                      | PASS                    |
| T09.7 | NN inference avg ≤ 30 ms over 6 images<br>avg=16.9ms worst=16.9ms n=6                                      | PASS                    |
| T10.1 | `camera awb auto` accepted                                                                                 | PASS                    |
| T10.2 | `camera exposure 10000` accepted                                                                           | PASS                    |
| T10.3 | `camera gain 0` accepted                                                                                   | PASS                    |
| T11.1 | 1_person.jpg (single) → ≥1 person<br>detected: person=1 vehicle=0 (gt 1p/0v) NN=16.8ms                     | PASS                    |
| T11.2 | 2_people.jpg (small group) → ≥1 person<br>detected: person=1 vehicle=0 (gt 2p/0v) NN=16.8ms                | PASS                    |
| T11.3 | 3_people.jpg (small group) → ≥2 person<br>detected: person=3 vehicle=0 (gt 3p/0v) NN=16.8ms                | PASS                    |
| T11.4 | 5_people.jpg (group) → ≥2 person<br>detected: person=0 vehicle=0 (gt 5p/0v) NN=16.8ms                      | FAIL —<br>person<br>0<2 |
| T11.5 | 7_people.jpg (group) → ≥2 person<br>detected: person=4 vehicle=0 (gt 7p/0v) NN=16.8ms                      | PASS                    |
| T11.6 | crowd_13.jpg (crowd) → ≥3 person<br>detected: person=0 vehicle=0 (gt 13p/0v) NN=16.8ms                     | FAIL —<br>person<br>0<3 |
| T11.7 | cars_18.jpg (vehicle (multi-class only)) → ≥0 person<br>detected: person=0 vehicle=0 (gt 0p/18v) NN=16.8ms | PASS                    |
| T11.8 | cars_15.jpg (vehicle (multi-class only)) → ≥0 person<br>detected: person=0 vehicle=0 (gt 0p/15v) NN=16.8ms | PASS                    |
| T11.9 | 13ppl_5trucks.jpg (mixed) → ≥3 person<br>detected: person=4 vehicle=0 (gt 13p/5v) NN=16.8ms                | PASS                    |

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